

SCSI Pass-Through (SPT)
2019 Update (TOI)

by

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Agenda

- What is SPT?
- Why use SPT?
- SPT Features
- New Features
- SPT Roadmap

SCSI Pass-Through (SPT) Utility

- SPT allows SCSI testing via the SCSI pass-through API
 - This is an OS specific user level API for issuing SCSI operations.
- Supported on most OS's, except OSX (lacks SCSI API)
- A fair amount of code originated from [DT tool](#)
- SPT is open source, and is available on GitHub:
 - <https://github.com/RobinTMiller/spt>
 - Note: Github does NOT include Vendor Unique (VU) extensions.
- Storage Enclosure Services (SES) support

Why Use This SPT Tool?

- Advanced Testing Requires Advanced Test Tools
- Key SPT Features: (one tool versus many)
 - Specifically designed for SCSI testing
 - Supports multiple jobs/threads for high load & stress
 - Supports JavaScript Object Notation (JSON)
- Useful for developers, automation, and support.
- This TOI will hopefully whet your appetite!
- We have internal developer (Robin) and support! 😊

SPT Features

- Send/Receive Diagnostic Support (added for SES pages)
- Support for I/O Ranges (for read/write/verify operations)
- Support for I/O Modes (copy/verify/mirror operations)
- Support for Multiple Slices (threads for higher load & stress)
- Support for Extended Copy Operations (two methods).
- Triage/Recovery Support (detailed errors, recovery, etc)
- Extended SCSI CDB Support (encode/decode CDB's)
- Enhanced Script File Support (custom test scripts)
- Emit Status Control Strings (roll your own output format)
- Test Check / Expect Options (automatic data/sense checks)
- Provides ASCII and JSON output formats! (work-in-progress)
- Support to unpack SCSI data (decode data in user format)

New SPT Version 3.x Features

- SES LED Control
- Log Pages: counters, temperature
- Built-in Request Sense parsing, and others
- Show Devices Formats:
 - Summary Format
 - Long Format
 - Roll-Your-Own
 - JSON Format
- List Disks/Enclosures Script: *lsdisks* (Python spt wrapper)
- Test Commands: selftest, extended self test, etc
- Python Shared Library via CTYPES API

SCSI XCOPY Acronyms

- Extended Copy (**XCOPY**) (older method used by Vmware)
- Token Based XCOPY:
 - Newer method used by Windows, aka [ODX](#)
 - Populate Token (**PT**)
 - Receive ROD Token Information (**RTTI**)
 - ROD = Representation of Data
 - Write Using Token (**WUT**)
 - **Token** - A Token represents point-in-time copy of LBA ranges from source LUN

What Are (some of) The Differences?

- Non-Token Based XCOPY: (VMware)
 - One SCSI Command:
 - Extended Copy CDB (XCOPY)
 - Inquiry VPD Device ID Identifier required
 - Multiple Sources, One Destination LUN
- Token Based XCOPY: (Microsoft)
 - Multiple SCSI Commands:
 - PT / RTTI / WUT / RTTI
 - One Source and One Destination LUN
 - Token XCOPY Volume Requirements

SCSI Pass-Through (SPT) Support

- Supports Both XCOPY Formats:
 - cdb=0x83 initiates non-token xcopy
 - cdb='0x83 0x11' initiates token xcopy
- Multiple Sources for non-token xcopy
- Multiple Destination for token xcopy
- Zero ROD Supported for Token xcopy
 - same as unmapping blocks via Unmap CDB
- Extended Debug (enable=xdebug)
 - automatically dumped during errors

Preparing Disks for Extended Copy Testing

```
# spt dsf=/dev/mapper/mpatha cdb=8e bs=32k starting=0 pattern=iot slices=10 enable=Debug
spt (j:0 t:0): Device: /dev/mapper/mpatha, Device Size: 512 bytes, Capacity: 20971520 blocks (thread 0)
spt (j:0 t:0): Device: /dev/mapper/mpatha, Starting lba=0, ending lba=20971519, block limit=20971520 (thread 0)
spt (j:1 t:2): Device: /dev/mapper/mpatha, Starting lba=2097152, ending lba=4194303, block limit=2097152 (thread 2)
spt (j:1 t:1): Device: /dev/mapper/mpatha, Starting lba=0, ending lba=2097151, block limit=2097152 (thread 1)
spt (j:1 t:3): Device: /dev/mapper/mpatha, Starting lba=4194304, ending lba=6291455, block limit=2097152 (thread 3)
spt (j:1 t:4): Device: /dev/mapper/mpatha, Starting lba=6291456, ending lba=8388607, block limit=2097152 (thread 4)
spt (j:1 t:8): Device: /dev/mapper/mpatha, Starting lba=14680064, ending lba=16777215, block limit=2097152 (thread 8)
spt (j:1 t:6): Device: /dev/mapper/mpatha, Starting lba=10485760, ending lba=12582911, block limit=2097152 (thread 6)
spt (j:1 t:5): Device: /dev/mapper/mpatha, Starting lba=8388608, ending lba=10485759, block limit=2097152 (thread 5)
spt (j:1 t:7): Device: /dev/mapper/mpatha, Starting lba=12582912, ending lba=14680063, block limit=2097152 (thread 7)
spt (j:1 t:9): Device: /dev/mapper/mpatha, Starting lba=16777216, ending lba=18874367, block limit=2097152 (thread 9)
spt (j:1 t:10): Device: /dev/mapper/mpatha, Starting lba=18874368, ending lba=20971519, block limit=2097152 (thread 10)
# spt dsf=/dev/mapper/mpatha cdb='9e 12' starting=0 logprefix=
```

Get LBA Status Information:

```
    Mapped Blocks: 20971520
    Deallocated Blocks: 0
    Total Blocks: 20971520
```

```
# spt dsf=/dev/mapper/mpathb cdb='42' starting=0 logprefix=
# spt dsf=/dev/mapper/mpathb cdb='9e 12' starting=0 logprefix=
```

Get LBA Status Information:

```
    Mapped Blocks: 0
    Deallocated Blocks: 20971520
    Total Blocks: 20971520
```

#

Non-Token Based XCOPY

```
# spt cdb=0x83 enable=recovery,sense src=/dev/mapper/mpatha dst=/dev/mapper/mpathb emit=multi logprefix=
```

```
    Thread: 1
    Source Device: /dev/mapper/mpatha
    Source Info: Block Length=512, Capacity=20971520
    Block Range: 0 - 20971520
Destination Device: /dev/mapper/mpathb
Destination Info: Block Length=512, Capacity=20971520
    Block Range: 0 - 20971520
    SCSI Name: Extended Copy
    SCSI CDB: 83 00 00 00 00 00 00 00 00 00 00 00 00 00 6c 00 00
    Data Direction: write
    Data Length: 512
    Exit Status: 0 = SUCCESS
    Host Status: 0 = DID_OK
    Driver Status: 0 = DRIVER_OK
    SCSI Status: 0 = SCSI_GOOD
    Sense Code: 0 = NO_CODE
    Sense Key: 0 = NO_SENSE
    asc/asq: (00, 00) = No additional sense information
Bytes Transferred: 108 (data bytes transferred)
    Residual: 0 (bytes not transferred)
    Iterations: 1
    Total Bytes: 276480
    Total Blocks: 20971520
Total Operations: 2564
    Sense Data: 00 00 00 00 00 00 00 00
Elapsed Time: 00m12.90s
    Starting Time: Fri Feb  1 14:27:50 2019
    Ending Time:  Fri Feb  1 14:28:03 2019
```

```
#
```

Token Based XCOPY Example

```
# spt cdb='0x83 0x11' enable=recovery,sense src=/dev/mapper/mpatha dst=/dev/mapper/mpathb emit=multi logprefix=
```

```
    Thread: 1
    Source Device: /dev/mapper/mpatha
    Source Info: Block Length=512, Capacity=20971520
    Block Range: 0 - 20971520
Destination Device: /dev/mapper/mpathb
Destination Info: Block Length=512, Capacity=20971520
    Block Range: 0 - 20971520
    SCSI Name: XCOPY - Write Using Token
    SCSI CDB: 83 11 00 00 00 00 01 8b 46 06 00 00 02 28 00 00
    Data Direction: write
    Data Length: 512
    Exit Status: 0 = SUCCESS
    Host Status: 0 = DID_OK
    Driver Status: 0 = DRIVER_OK
    SCSI Status: 0 = SCSI_GOOD
    Sense Code: 0 = NO_CODE
    Sense Key: 0 = NO_SENSE
    asc/asq: (00, 00) = No additional sense information
Bytes Transferred: 32 (data bytes transferred)
    Residual: 773 (bytes not transferred)
    Iterations: 1
    Total Bytes: 88320
    Total Blocks: 20971520
Total Operations: 644
    Sense Data: 00 00 00 00 00 00 00 00
Elapsed Time: 00m10.60s
    Starting Time: Fri Feb 1 14:29:50 2019
    Ending Time: Fri Feb 1 14:30:01 2019
```

```
#
```

Verifying the XCOPY Operation

```
# spt iomode=verify length=32k dsf=${SRC} starting=0 dsf1=${DST} starting=0 emit=multi enable=compare,recovery,sense logprefix=
```

```
    Thread: 1
    Source Device: /dev/mapper/mpathb
    Source Info: Block Length=512, Capacity=20971520
    Block Range: 0 - 20971520
Destination Device: /dev/mapper/mpatha
Destination Info: Block Length=512, Capacity=20971520
    Block Range: 0 - 20971520
    SCSI Name: Read(16)
    SCSI CDB: 88 00 00 00 00 00 01 3f ff c0 00 00 00 40 00 00
Data Direction: read
Data Length: 32768
Exit Status: 0 = SUCCESS
Host Status: 0 = DID_OK
Driver Status: 0 = DRIVER_OK
SCSI Status: 0 = SCSI_GOOD
Sense Code: 0 = NO_CODE
Sense Key: 0 = NO_SENSE
    asc/asq: (00, 00) = No additional sense information
Bytes Transferred: 32768 (data bytes transferred)
    Residual: 0 (bytes not transferred)
    Iterations: 1
    Total Bytes: 21474836480
    Total Blocks: 41943040
Total Operations: 655362
    Sense Data: 00 00 00 00 00 00 00 00
Elapsed Time: 01m35.04s
Starting Time: Fri Feb 1 14:13:29 2019
Ending Time: Fri Feb 1 14:15:04 2019
```

```
#
```

Alternate Data Verification with dt Tool

```
# dt src=${SRC} dst=${DST} iomode=verify bs=32k slices=10 workload=job_stats_only logprefix= disable=scsi
```

Operating System Information:

```
Host name: rtpqa-rhel001 (10.64.21.59)
User name: romiller
Process ID: 28933
OS information: Linux 3.10.0-514.el7.x86_64 #1 SMP Wed Oct 19 11:24:13 EDT 2016 x86_64
```

Job Statistics:

```
Input device/file name: /dev/mapper/mpatha (device type=disk)
Type of I/O's performed: sequential (forward)
Job Information Reported: Job 1
Number of slices: 10
Data operation performed: Verified '/dev/mapper/mpatha' with '/dev/mapper/mpathb'.
Total records read: 655360
Total bytes read: 21474836480 (20971520.000 Kbytes, 20480.000 Mbytes, 20.000 Gbytes)
Total records written: 0
Total bytes written: 0 (0.000 Kbytes, 0.000 Mbytes, 0.000 Gbytes)
Total records processed: 655360 @ 32768 bytes/record (32.000 Kbytes)
Total bytes transferred: 21474836480 (20971520.000 Kbytes, 20480.000 Mbytes)
Average transfer rates: 733494714 bytes/sec, 716303.432 Kbytes/sec
Number I/O's per second: 22384.482
Number seconds per I/O: 0.0000 (0.04ms)
Total passes completed: 1/1
Total errors detected: 0/1
Total elapsed time: 00m29.28s
Total system time: 00m10.45s
Total user time: 00m42.31s
Starting time: Fri Feb 1 14:37:33 2019
Ending time: Fri Feb 1 14:38:03 2019
```

```
#
```

Zero ROD Token to Unmap Blocks

```
# spt dsf=${DST} cdb='9e 12' starting=0 logprefix=
```

```
Get LBA Status Information:
```

```
      Mapped Blocks: 20971520
Deallocated Blocks: 0
      Total Blocks: 20971520
```

```
# spt cdb='83 11' dsf=${DST} starting=0 enable=zerorod enable=recovery,sense emit=default logprefix=
```

```
      Thread: 1
      Device Name: /dev/mapper/mpathb
      Device Info: Block Length=512, Capacity=20971520
      Block Range: 0 - 20971520
      SCSI Name: XCOPY - Write Using Token
      SCSI CDB: 83 11 00 00 00 00 01 8b 46 06 00 00 02 28 00 00
Data Direction: write
Data Length: 512
Exit Status: 0 = SUCCESS
Host Status: 0 = DID_OK
Driver Status: 0 = DRIVER_OK
SCSI Status: 0 = SCSI_GOOD
Sense Code: 0 = NO_CODE
Sense Key: 0 = NO_SENSE
      asc/asq: (00, 00) = No additional sense information
Bytes Transferred: 32 (data bytes transferred)
Residual: 773 (bytes not transferred)
Iterations: 1
Total Bytes: 88320
Total Blocks: 20971520
Total Operations: 322
Sense Data: 00 00 00 00 00 00 00 00
      Elapsed Time: 00m00.19s
Starting Time: Fri Feb 1 14:43:29 2019
Ending Time: Fri Feb 1 14:43:30 2019
```

```
# spt dsf=${DST} cdb='9e 12' starting=0 logprefix=
```

```
Get LBA Status Information:
```

```
      Mapped Blocks: 0
Deallocated Blocks: 20971520
      Total Blocks: 20971520
```

```
#
```

SES LED Control (useful on array)

- Modify SES Control Elements
- Format:
spt ses {clear|set}={devoff|fail/fault|ident/locate|unlock}
- Examples:

```
# export SPT_DEVICE=/dev/sg50
# spt ses set=ident etype=array element=85
# spt logprefix= rcvdiag etype=array element=85 page=enclosure
```

Enclosure Status Diagnostic Page:

Page Code: 0x2

...

Element Descriptor: 85

Element Text: Array Slots

Element Status Code: OK (0x01)

...

Identify LED: On

...

```
# spt ses clear=ident etype=array element=85
```

Log Pages

- Format:

spt logsense page={name|code}

- Examples:

spt logsense page=write_error logprefix= rfmt=brief

Write Error Counter Parameters (Page 0x2 - Current Cumulative Values):

Errors Corrected w/o Substantial Delay: 0

Errors Corrected with Possible Delays: 8

Total re-Reads or re-Writes: 0

Total Errors Corrected: 8

Times Correction Algorithm Processed: 17040

Total Bytes Processed: 3097992679424

Total Uncorrected Errors: 0

#

Request Sense

- Provide Request Sense for In-Progress Commands
- Format:

spt requestsense

- Examples:

```
# spt test bextended
# spt requestsense logprefix=
```

Request Sense Data: (sense length 32 bytes)

```

                                Error Code: 0x70 = Current Error
Information Field Valid: No
                                Sense Key: 0x2 = NOT READY
Information Field: 0 (0)
Additional Sense Length: 24 (0x18)
Command Specific Information: 0 (0)
Additional Sense Code/Qualifier: (0x4, 0x9) - Logical unit not ready, self-test in progress
Field Replaceable Unit Code: 0
Sense Key Specific Bytes: 80 03 08
Progress Indication: 1.18%
Additional Sense Bytes: 00 00 f5 05 00 00 00 00 00 00 00 00 00 00 00
```

```
#
```

Show Device Information

- Show SCSI device information
- Format:

`# spt show { devices | edt } [filters]`

- Example:

Show Devices Filters:

<code>device(s)=string</code>	The device path(s).
<code>device_type(s)={value string},...</code>	The device type. (or dtype=)
<code>exclude=string</code>	The exclude path(s).
<code>product=string</code>	The product name. (or pid=)
<code>vendor=string</code>	The vendor name. (or vid=)
<code>revision=string</code>	The revision level. (or rev=)
<code>fw_version=string</code>	The firmware version. (or fwver=)
<code>serial=string</code>	The serial number.
<code>show-fields=string</code>	Show devices brief fields. (or sflds=).
<code>show-format=string</code>	Show devices format control. (or sfmt=).
<code>show-path=string,...</code>	Show devices using path. (or spath=).

Show Devices Example

```
# lsdisks --vendor Nimble
```

Linux Device	SCSI Device	Vendor	Product	Array SW Version	Drive Capacity	Block Length	Serial Number
/dev/sdj	/dev/sg10	Nimble	Server	5.1	20971520	512	92586db90a929ca96c9ce9004d05262d
/dev/sdl	/dev/sg12	Nimble	Server	5.1	20971520	512	92586db90a929ca96c9ce9004d05262d
/dev/sdn	/dev/sg14	Nimble	Server	5.1	20971520	512	92586db90a929ca96c9ce9004d05262d
/dev/sdp	/dev/sg16	Nimble	Server	5.1	20971520	512	92586db90a929ca96c9ce9004d05262d
...							
/dev/sdi	/dev/sg9	Nimble	Server	5.1	20971520	512	7de17fd7510524536c9ce9004d05262d
/dev/sds	/dev/sg19	Nimble	Server	5.1	20971520	512	7de17fd7510524536c9ce9004d05262d
/dev/sdu	/dev/sg21	Nimble	Server	5.1	20971520	512	7de17fd7510524536c9ce9004d05262d
/dev/sdw	/dev/sg23	Nimble	Server	5.1	20971520	512	7de17fd7510524536c9ce9004d05262d
/dev/sdy	/dev/sg25	Nimble	Server	5.1	20971520	512	7de17fd7510524536c9ce9004d05262d

```
#
```

Show Devices Fields Example

```
# spt show devices vid=Nimble show-fields='vid,serial,swver,syncrep,paths'
```

Vendor	Serial Number	SW Version	Sync Rep	Device Paths
Nimble	1ab3574fff125ce66c9ce9004d05262d	5.1	False	/dev/sde /dev/sg5 /dev/sdg /dev/sg7 /dev/mapper/mpathd /dev/mapper/21ab3574fff125ce66c9ce9004d05262d
Nimble	34b7a7a09b7abc1e6c9ce9004d05262d	5.1	False	/dev/sdf /dev/sg6 /dev/sdh /dev/sg8 /dev/mapper/mpathe /dev/mapper/234b7a7a09b7abc1e6c9ce9004d05262d
Nimble	eb17d3b14c2f1bf36c9ce9004d05262d	5.1	False	/dev/sdi /dev/sg9 /dev/sdk /dev/sg11 /dev/mapper/mpathf /dev/mapper/2eb17d3b14c2f1bf36c9ce9004d05262d
Nimble	05eaa367468617896c9ce9004d05262d	5.1	False	/dev/sdj /dev/sg10 /dev/sdl /dev/sg12 /dev/mapper/mpathg /dev/mapper/205eaa367468617896c9ce9004d05262d
Nimble	f081af3d26bad6296c9ce9004d05262d	5.1	False	/dev/sdc /dev/sg3 /dev/sdn /dev/sg14 /dev/mapper/mpathb /dev/mapper/2f081af3d26bad6296c9ce9004d05262d
Nimble	c16c750ad1c5ff9a6c9ce9004d05262d	5.1	False	/dev/sdd /dev/sg4 /dev/sdo /dev/sg15 /dev/mapper/mpathc /dev/mapper/2c16c750ad1c5ff9a6c9ce9004d05262d
Nimble	d68d651c8052829e6c9ce9004d05262d	5.1	False	/dev/sdb /dev/sg2 /dev/sdp /dev/sg16 /dev/mapper/mpatha /dev/mapper/2d68d651c8052829e6c9ce9004d05262d
Nimble	a7e2da4ed76e35fe6c9ce9004d05262d	5.1	False	/dev/sdm /dev/sg13 /dev/sdq /dev/sg17 /dev/mapper/mpathh /dev/mapper/2a7e2da4ed76e35fe6c9ce9004d05262d

```
#
```

Show Device JSON Output Example

```
# spt show edt device=/dev/mapper/mpathh ofmt=json
{
  "SCSI Devices": {
    "Device List": [
      {
        "Peripheral Device Type": 0,
        "Peripheral Device Type Description": "Direct Access Device",
        "Vendor Identification": "Nimble ",
        "Product Identification": "Server ",
        "Firmware Revision Level": "1.0 ",
        "Full Firmware Version": "<not available>",
        "Product Serial Number": "a7e2da4ed76e35fe6c9ce9004d05262d",
        "Device World Wide Name": "0xa7e2da4ed76e35fe6c9ce9004d05262d",
        "Device Target Port": "<not available>",
        "Array SW Version": "5.1",
        "Target Type": "Volume Scoped",
        "Sync Replication": false,
        "Device Paths": "\/dev\/mapper\/mpathh",
        "Path Types": [
          {
            "DMMP Device": "\/dev\/mapper\/mpathh",
            "SCSI Nexus": "[27:0:0:0]"
          }
        ]
      }
    ],
    "JSON Status": 0
  }
}
```

Show SCSI Sense Information

- Format:

spt show scsi [filters]

- Examples:

spt show scsi ascq=0x0404

Sense Code/Qualifier = (0x4, 0x4) = Logical unit not ready, format in progress

spt show scsi key=0x2

Sense Key = 0x2 = NOT READY

spt show scsi status=28

SCSI Status = 0x28 = SCSI_QUEUE_FULL

#

List Disks Script

- Python Script Utilizing spt to List Disks/Enclosure Information
- Format:

lsdisks [options...]

- Help Text:

```
# lsdisks --help
usage: lsdisks [-h] [--debug] [--drives DRIVES] [--exclude EXCLUDE] [--force_spt] [--json] [--long] [--noencs] [--noheader]
               [--power_on_hours] [--firmware_version FIRMWARE_VERSION] [--product_name PRODUCT_NAME] [--vendor_name VENDOR_NAME]
               [--serial_number SERIAL_NUMBER] [--sas_address SAS_ADDRESS] [--target_port TARGET_PORT] [--use_lsscsi]
               [--spt_path SPT_PATH]

optional arguments:
  -h, --help                show this help message and exit
  --debug                   The debug flag. (Default: False)
  --drives DRIVES           The drives to display. (Default: [])
  --exclude EXCLUDE        The drives to exclude. (Default: [])
  --force_spt              Force using spt (debug). (Default: False)
  --json                   Enable JSON format. (Default: False)
  --long                   Enable long format. (Default: False)
  --noencs                 Exclude enclosures. (Default: False)
  --noheader               Exclude headers. (Default: False)
  --power_on_hours          Include power on hours. (Default: True)
  --firmware_version FIRMWARE_VERSION
                           The firmware version. (Default: None)
  --product_name PRODUCT_NAME
                           The product name. (Default: None)
  --vendor_name VENDOR_NAME
                           The vendor name. (Default: None)
  --serial_number SERIAL_NUMBER
                           The serial number. (Default: None)
  --sas_address SAS_ADDRESS
                           The SAS address. (Default: None)
  --target_port TARGET_PORT
                           The target port. (Default: None)
  --use_lsscsi             Find devices via lsscsi. (Default: False)
  --spt_path SPT_PATH      The spt tool path. (Default: /usr/local/bin/spt)
#
```

List Disks Example

```
# lsdisks
```

Linux Device	SCSI Device	Vendor	Product	Array SW Version	Drive Capacity	Block Length	Serial Number
/dev/sda	/dev/sg0	VMware	Virtual disk		83886080	512	<not available>
/dev/sde	/dev/sg5	Nimble	Server	5.1	20971520	512	1ab3574ffff125ce66c9ce9004d05262d
/dev/sdg	/dev/sg7	Nimble	Server	5.1	20971520	512	1ab3574ffff125ce66c9ce9004d05262d
/dev/sdf	/dev/sg6	Nimble	Server	5.1	20971520	512	34b7a7a09b7abc1e6c9ce9004d05262d
/dev/sdh	/dev/sg8	Nimble	Server	5.1	20971520	512	34b7a7a09b7abc1e6c9ce9004d05262d
/dev/sdi	/dev/sg9	Nimble	Server	5.1	20971520	512	eb17d3b14c2f1bf36c9ce9004d05262d
/dev/sdk	/dev/sg11	Nimble	Server	5.1	20971520	512	eb17d3b14c2f1bf36c9ce9004d05262d
/dev/sdj	/dev/sg10	Nimble	Server	5.1	20971520	512	05eaa367468617896c9ce9004d05262d
/dev/sdl	/dev/sg12	Nimble	Server	5.1	20971520	512	05eaa367468617896c9ce9004d05262d
/dev/sdc	/dev/sg3	Nimble	Server	5.1	20971520	512	f081af3d26bad6296c9ce9004d05262d
/dev/sdn	/dev/sg14	Nimble	Server	5.1	20971520	512	f081af3d26bad6296c9ce9004d05262d
/dev/sdd	/dev/sg4	Nimble	Server	5.1	20971520	512	c16c750ad1c5ff9a6c9ce9004d05262d
/dev/sdo	/dev/sg15	Nimble	Server	5.1	20971520	512	c16c750ad1c5ff9a6c9ce9004d05262d
/dev/sdb	/dev/sg2	Nimble	Server	5.1	20971520	512	d68d651c8052829e6c9ce9004d05262d
/dev/sdp	/dev/sg16	Nimble	Server	5.1	20971520	512	d68d651c8052829e6c9ce9004d05262d
/dev/sdm	/dev/sg13	Nimble	Server	5.1	20971520	512	a7e2da4ed76e35fe6c9ce9004d05262d
/dev/sdq	/dev/sg17	Nimble	Server	5.1	20971520	512	a7e2da4ed76e35fe6c9ce9004d05262d

```
#
```

List Disks Long Format Example

```
# lsdisks --long --drives=/dev/sdq
```

```
        Device Type: disk
    Device Description: Direct Access Device
    Linux Device Name: /dev/sdq
        SCSI Device Name: /dev/sg17
Product Identification: Server
    Vendor Identification: Nimble
        Firmware Version: 1.0
        Array SW Version: 5.1
            Serial Number: a7e2da4ed76e35fe6c9ce9004d05262d
        Drive Capacity: 20971520
            Block Length: 512
```

```
#
```

List Disks JSON Format Example

```
# lsdisks --long --drives=/dev/sdq --json
[
  {
    "SAS Address": "",
    "DMMP Device Name": "",
    "SCSI Device Name": "/dev/sg17",
    "Product Identification": "Server",
    "Drive Capacity": 20971520,
    "Linux Device Name": "/dev/sdq",
    "Device World Wide Name": "0xa7e2da4ed76e35fe6c9ce9004d05262d",
    "Power On Hours": "",
    "Device Type": "disk",
    "Serial Number": "a7e2da4ed76e35fe6c9ce9004d05262d",
    "Device Type Description": "Direct Access Device",
    "Block Length": 512,
    "Enclosure Slot": "",
    "Firmware Version": "1.0",
    "Enclosure Device": "",
    "Vendor Identification": "Nimble",
    "Revision Level": "1.0",
    "Slot Description": "",
    "Linux SCSI Nexus": "[27:0:0:0]",
    "Current Temperature": "",
    "Array SW Version": "5.1"
  }
]
#
```

List Disks on Nimble Array

```
rtp-afa77-A:tmp# ./lsdisks
```

Linux Device	SCSI Device	Vendor	Product	Firmware Revision	Drive Capacity	Block Length	Curr Temp	Serial Number	SAS Address	Enc SCSI Device	Enc Slot	Enc Slot Description
/dev/sda	/dev/sg1	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	23 C	PHDV7473006T480BGN	0x56c9ce9f002da2b9	/dev/sg48	0	ArrayDevice00
/dev/sdb	/dev/sg3	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV749404B5480BGN	0x56c9ce9f002da3f9	/dev/sg48	4	ArrayDevice04
/dev/sdc	/dev/sg5	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	23 C	PHDV7473002W480BGN	0x56c9ce9f002da971	/dev/sg48	8	ArrayDevice08
/dev/sdd	/dev/sg7	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV745100A5480BGN	0x56c9ce9f002dad1	/dev/sg48	12	ArrayDevice0C
/dev/sde	/dev/sg9	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV749404F1480BGN	0x56c9ce9f002daa09	/dev/sg48	16	ArrayDevice10
/dev/sdf	/dev/sg11	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV745000YC480BGN	0x56c9ce9f002da749	/dev/sg48	20	ArrayDevice14
/dev/sdg	/dev/sg13	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV7451003C480BGN	0x56c9ce9f002dae11	/dev/sg48	1	ArrayDevice01
/dev/sdh	/dev/sg15	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	23 C	PHDV74500195480BGN	0x56c9ce9f002da351	/dev/sg48	5	ArrayDevice05
/dev/sdi	/dev/sg17	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	23 C	PHDV74510140480BGN	0x56c9ce9f002da851	/dev/sg48	9	ArrayDevice09
/dev/sdj	/dev/sg19	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV74510038480BGN	0x56c9ce9f002daa19	/dev/sg48	13	ArrayDevice0D
/dev/sdk	/dev/sg21	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV745000YS480BGN	0x56c9ce9f002da7e9	/dev/sg48	17	ArrayDevice11
/dev/sdl	/dev/sg23	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV745101J0480BGN	0x56c9ce9f002da9c9	/dev/sg48	21	ArrayDevice15
/dev/sdm	/dev/sg25	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV7450014Q480BGN	0x56c9ce9f002db109	/dev/sg48	2	ArrayDevice02
/dev/sdn	/dev/sg27	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV747300ET480BGN	0x56c9ce9f002ddf49	/dev/sg48	6	ArrayDevice06
/dev/sdo	/dev/sg29	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	23 C	PHDV74730056480BGN	0x56c9ce9f002dada9	/dev/sg48	10	ArrayDevice0A
/dev/sdp	/dev/sg31	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV7450002M480BGN	0x56c9ce9f002db0a1	/dev/sg48	14	ArrayDevice0E
/dev/sdq	/dev/sg33	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV7473005C480BGN	0x56c9ce9f002da6c9	/dev/sg48	18	ArrayDevice12
/dev/sdr	/dev/sg35	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	23 C	PHDV7473004F480BGN	0x56c9ce9f002dab41	/dev/sg48	22	ArrayDevice16
/dev/sds	/dev/sg37	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV7450015B480BGN	0x56c9ce9f002dade9	/dev/sg48	3	ArrayDevice03
/dev/sdt	/dev/sg39	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV7450017T480BGN	0x56c9ce9f002ddf89	/dev/sg48	7	ArrayDevice07
/dev/sdu	/dev/sg41	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV74500168480BGN	0x56c9ce9f002dac49	/dev/sg48	11	ArrayDevice0B
/dev/sdv	/dev/sg43	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV7450002A480BGN	0x56c9ce9f002dac89	/dev/sg48	15	ArrayDevice0F
/dev/sdw	/dev/sg45	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV745000YN480BGN	0x56c9ce9f002da7b9	/dev/sg48	19	ArrayDevice13
/dev/sdx	/dev/sg47	ATA	INTEL SSDSC2BB48	N2010112	937703088	512	22 C	PHDV747300F4480BGN	0x56c9ce9f002db051	/dev/sg48	23	ArrayDevice17
/dev/sdy	/dev/sg49	ATA	SQF-SM4U2-16G-9N	S9FUA028	31277232	512		FEB40781040800295992				

```
rtp-afa77-A:tmp#
```

Test Commands

- Perform a Diagnostic Test
- Format:

spt test options

Test Options:

<code>abort</code>	Abort a background test.
<code>selftest</code>	Standard self test.
<code>bextended</code>	Background extended self test.
<code>bshort</code>	Background short self test.
<code>extended</code>	Foreground extended self test.
<code>short</code>	Foreground short self test.

- Examples:

```
# spt testself  
# spt test short
```

spt Python Shared Library

- Uses Python CTYPES to call spt shared library
- Snippet from spt unit test script (sptp/sptut.py):

```
# Note: This will be packaged with spt and copied to /usr/local/bin
self.libspt = CDLL('./libspt.so')
self.sptapi = self.libspt.spt
self.program = "sptut"
#
# int
# spt(char *stdin_buffer,
#      char *stderr_buffer, int stderr_length,
#      char *stdout_buffer, int stdout_length,
#      char *emit_status_buffer, int emit_status_length)
#
self.sptapi.restype = c_int
self.sptapi.argtypes = [c_char_p, c_char_p, c_int, c_char_p, c_int, c_char_p, c_int]
self.spt_args = {}
self.spt_args['stderr_length'] = STDERR_BUFFER_SIZE
self.spt_args['stdout_length'] = STDOUT_BUFFER_SIZE
self.spt_args['emit_status_length'] = EMIT_BUFFER_SIZE
self.spt_args['stderr_buffer'] = create_string_buffer('\000' * self.spt_args['stderr_length'])
self.spt_args['stdout_buffer'] = create_string_buffer('\000' * self.spt_args['stdout_length'])
self.spt_args['emit_status_buffer'] = create_string_buffer('\000' * self.spt_args['emit_status_length'])
self.spt_args['stdout_dict'] = None
self.spt_args['emit_dict'] = None
...
```

- sptut.py is Multi-Threaded Script calling Threaded spt API.

spt Shared Library Unit Test Script

```
# ./sptut.py --help
usage: sptut [-h] [--debug] [--verify]

optional arguments:
  -h, --help  show this help message and exit
  --debug     The debug flag. (Default: False)
  --verify    Do verify media. (Default: False)
# ./sptut.py --verify
Executing: version
Author: Robin T. Miller, Date: October 29, 2018, Version: 3.16
Executing: show devices dtype=direct
Starting thread for /dev/sda...
Executing: dsf=/dev/sda cdb=0 status=good retry=100 msleep=100 enable=wait
Starting thread for /dev/sde...
Executing: dsf=/dev/sde cdb=0 status=good retry=100 msleep=100 enable=wait
Starting thread for /dev/sdg...
Executing: dsf=/dev/sdg cdb=0 status=good retry=100 msleep=100 enable=wait
Starting thread for /dev/sdf...
Executing: dsf=/dev/sdf cdb=0 status=good retry=100 msleep=100 enable=wait
Executing: dsf=/dev/sdf inquiry
Executing: dsf=/dev/sdf readcapacity16
Executing: dsf=/dev/sdf inquiry page=serial
Executing: dsf=/dev/sdf cdb=0x8f bs=32k limit=1g threads=1
...
# spt showopcodes | fgrep -i verify | fgrep 0x8f
0x8f    0x00    none    yes     no      65536  Verify(16)
#
```

SPT Roadmap

- Finish adding SES diagnostic page support.
- Add more Inquiry, Log, and Mode pages.
- Add firmware download and Format support.
- Add JSON output format for other SCSI operations.
- Add SCSI compliance verification.
- Add support for Linux “bsg” SCSI interface.
- Add support for multiple devices. (like dt)
- Job Control for waiting on Async Threads. (like dt)
- Add server/client message passing using ZeroMQ.